

Predicting Visual Attention Transitions in Charts

A reproducible analysis of SalChartQA BubbleView sequences
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PROJECT SUMMARY

This project asks whether a simple sequential model can predict which chart region a viewer will inspect next while answering a question. It uses BubbleView mouse-click traces as a scalable proxy for visual attention and evaluates all models on images excluded from training. The analysis is intentionally baseline-driven: it tests how much predictive value comes from the current region, the question type, and the chart type before considering more complex models.

Dataset and evaluation	Value
Source	SalChartQA question-driven chart-attention dataset
Charts	3,000 images across horizontal bar, vertical bar, line, and pie charts
Responses used after sequence cleaning	73,829
Transition	825,310
Representation	6 x 6 spatial grid; repeated consecutive regions collapsed
Split	2,400 training images and 600 unseen test images, stratified by chart type

RESEARCH QUESTION

How accurately can the next attended chart region be predicted from the current region, and how much additional information is provided by the viewer's question type and the chart type?

The objective is not to reproduce eye tracking from mouse clicks or to claim a cognitive model of chart reading. It is to establish a transparent sequence-prediction baseline, quantify where conditioning helps, and identify the behavior that a stronger model would need to capture.

PROJECT CONTRIBUTION

Behavioral data: converts ordered attention-proxy traces into transitions rather than aggregating them into a single saliency map.

Computational behavior modeling: compares increasingly conditioned sequential models and evaluates generalization to unseen stimuli.

Controlled interpretation: uses ablation, image-level uncertainty, and failure analysis instead of relying on a headline accuracy alone.

Visualization: communicates model performance, heterogeneity, and systematic error patterns through interpretable figures.

DATA AND PREPROCESSING

SalChartQA contains question-driven BubbleView interactions for information visualizations. In BubbleView, a chart is blurred and each click reveals a local area. The click sequence provides an approximate record of where a participant chose to inspect the chart, but it is not a direct eye-tracking scanpath. The analysis therefore uses the term attention proxy throughout.

Each click coordinate was normalized to the displayed chart and assigned to one of 36 regions in a 6 x 6 grid. Invalid or out-of-bounds clicks were removed. Consecutive clicks in the same region were collapsed so that a

transition represents movement between regions rather than repeated inspection within one region. Sequences with fewer than two valid regions cannot produce a transition and were excluded from transition-level evaluation.

The split was performed by image rather than by transition. All responses for a chart remained in one partition, preventing the same visual stimulus from contributing transitions to both training and test data. The 80/20 split retained the chart-type distribution, resulting in 2,400 training images and 600 completely unseen test images.

MODELS

Model	Prediction rule	Purpose
Global frequency	Most frequent next region in training	Lower-bound behavioral baseline
Current-region Markov	Most frequent next region for each current region	Captures local spatial continuity
Question-conditioned	Conditions on current region and question type	Tests task-specific behavior
Chart-conditioned	Conditions on current region and chart type	Tests structure-specific behavior
Fully conditioned	Conditions on current region, question type and chart type	Tests their joint contribution

For exact condition combinations not observed during training, prediction falls back to the current-region transition distribution. Only six test cases required this fallback. Top-1 accuracy evaluates the most likely region; Top-3 accuracy treats a prediction as correct when the actual next region appears among the three most likely candidates.

EVALUATION

- ❖ Transition-level accuracy measures aggregate predictive performance across all test transitions.
- ❖ Per-image accuracy prevents charts with many transitions from dominating the interpretation.
- ❖ Ablation compares current-region, question-conditioned, chart-conditioned, and fully conditioned variants.
- ❖ An image-level bootstrap estimates uncertainty in the mean improvement over the current-region model.
- ❖ Error analysis compares actual and predicted spatial movement distance and identifies frequent transition confusions.

MAIN RESULTS

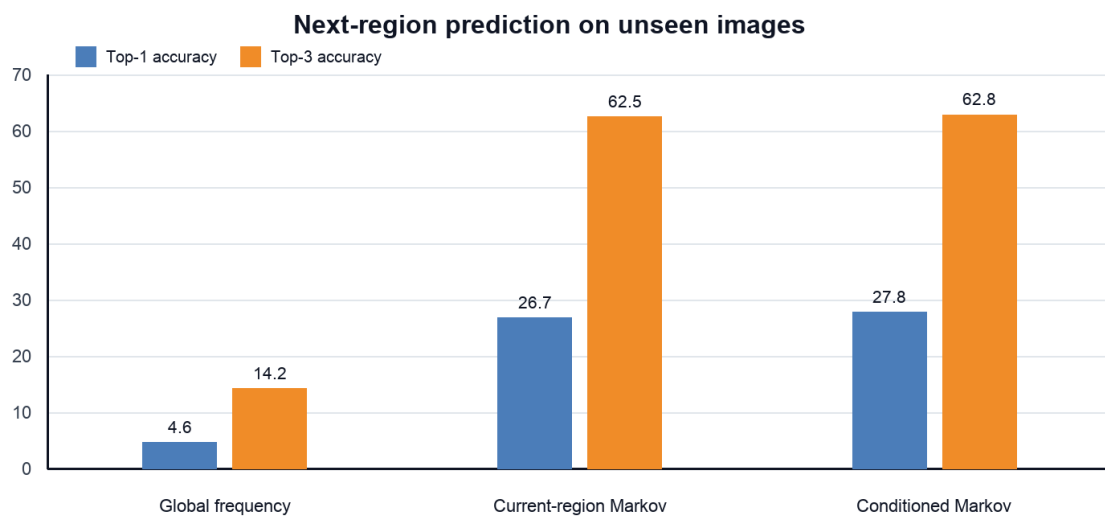


Figure 1 Top-1 and Top-3 accuracy on 600 unseen chart images.

Model	Top-1	Top-3	Gain over global Top-1
Global frequency	4.62%	14.16%	0.00 pp
Current-region Markov	26.74%	62.47%	+22.12 pp
Conditioned Markov	27.77%	62.84%	+23.14 pp

The current region contains most of the available predictive signal: moving from the global baseline to a first-order Markov model adds 22.12 percentage points in Top-1 accuracy. Adding question and chart context yields a smaller but consistent increase of 1.03 percentage points over the current-region model. The corresponding Top-3 gain is 0.37 percentage points.

IMAGE-LEVEL ROBUSTNESS

Across the 600 test images, the mean per-image accuracy was 24.91% for the current-region model and 26.22% for the conditioned model. Median accuracy increased from 25.37% to 26.81%. The conditioned model improved 357 images, left 25 unchanged, and worsened 218. The mean image-level gain was 1.31 percentage points, with a 95% bootstrap confidence interval of 0.90 to 1.72 percentage points.

ABLATION

Feature set	Top-1 accuracy	Gain over global Top-1
Current region	26.74%	0.00 pp
Current-region and question type	26.75%	+0.01 pp
Current-region and chart type	27.74%	+1.00 pp
Current-region, question type and chart type	27.77%	+1.03 pp

Chart type accounts for nearly all of the aggregate improvement. Question type adds almost nothing once current region and chart type are known. This result cautions against assuming that a semantically richer model is automatically more useful.

WHERE CONDITIONING HELPS AND HURTS

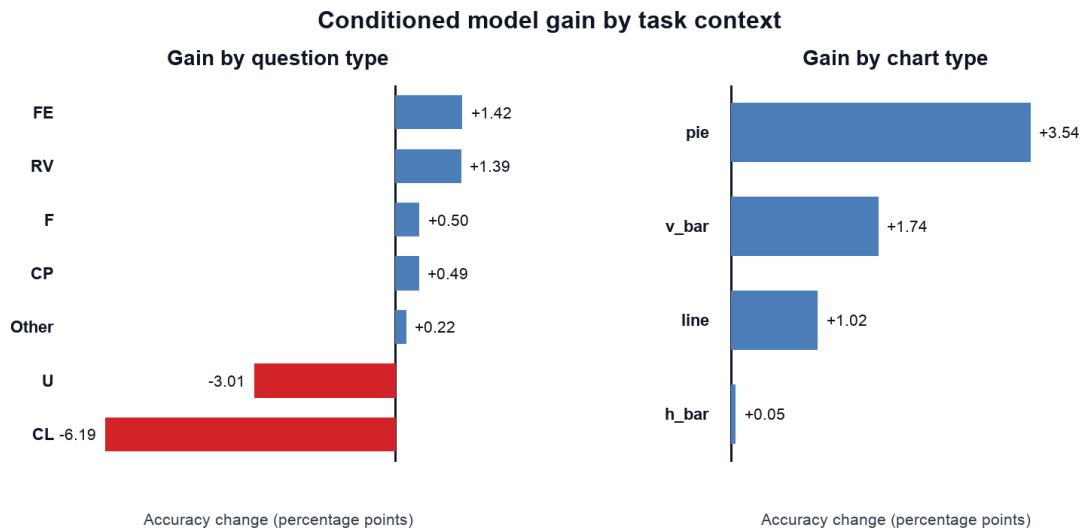


Figure 2 Conditioned-model gain by question type and chart type.

The conditioned model benefits pie charts most (+3.54 pp), followed by vertical bar charts (+1.74 pp) and line charts (+1.02 pp). Horizontal bars show almost no change (+0.05 pp). At question level, finding-extremum and retrieving-value questions gain about 1.4 pp, whereas U and CL questions decline by 3.01 and 6.19 pp respectively. The results therefore support targeted rather than universal conditioning.

FAILURE ANALYSIS

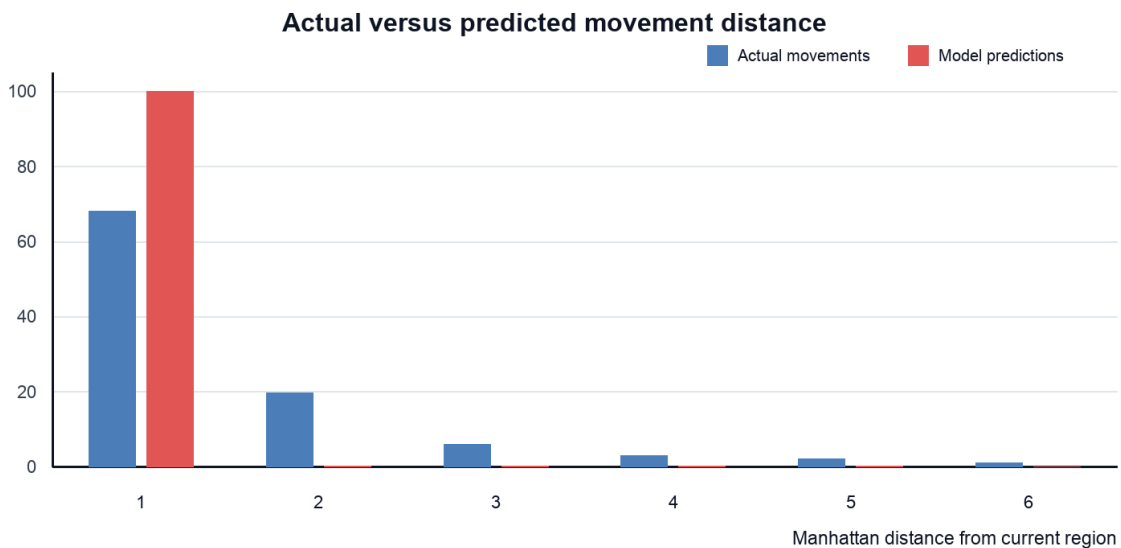


Figure 3 Distribution of actual and predicted Manhattan movement distance.

The principal limitation is not hidden by the aggregate accuracy: 67.98% of actual movements have Manhattan distance one, but only 99.90% of model predictions do. The model has essentially learned a local-neighbor heuristic. It achieves 40.83% accuracy for true distance-one movements but almost zero accuracy once the actual movement spans two or more grid cells. Frequent errors include predicting the neighboring region on the opposite side of the current location.

INTERPRETATION

The analysis provides three conclusions. First, spatial continuity is a strong baseline for chart-viewing sequences. Second, chart structure offers a modest and statistically stable improvement, whereas question type contributes little in this representation. Third, a first-order frequency model cannot represent long-range or semantically directed attention shifts. The small average gain should therefore be viewed as evidence for context dependence, not as a complete model of chart reading.

LIMITATIONS

- ❖ BubbleView clicks approximate visual attention but are not eye-tracker fixations. Clicking introduces a deliberate motor action and may omit covert or brief visual inspection.
- ❖ The 6 x 6 grid is interpretable but coarse. It ignores chart elements, labels, axes, and semantic relationships between regions.
- ❖ A first-order Markov model uses only the immediately preceding region and cannot represent longer histories, revisits, or task progress.
- ❖ The evaluation holds out images, not participants. It supports stimulus generalization but does not isolate participant-independent strategy generalization.
- ❖ Transition-level accuracy is dominated by common local movements. The per-image bootstrap and distance analysis reduce but do not remove this imbalance.

NEXT RESEARCH STEPS

- ❖ Model longer action histories using variable-order Markov models, recurrent architectures, or transformer-based sequence models.
- ❖ Use hierarchical evaluation that separates participant, image, chart type, and question effects.
- ❖ Compare supervised sequence prediction with computational-rationality or reinforcement-learning models that optimize information gain and task success.
- ❖ Build an interactive visual-analytics interface for inspecting trajectories, uncertainty, subgroup differences, and model failures.

REPRODUCIBILITY

The analysis is implemented in Python with pandas, NumPy, and matplotlib. The public release includes the executed notebook, an HTML export, fixed train and test image lists, summary CSV files, and figures so the results can be inspected without executing the code. The original SalChartQA data are not redistributed.

DATASET CITATION

Wang, Y., Wang, W., Abdelhafez, A., Elfares, M., Hu, Z., Bace, M., and Bulling, A. 2024. SalChartQA: Question-driven Saliency on Information Visualisations. Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems. <https://doi.org/10.1145/3613904.3642942>